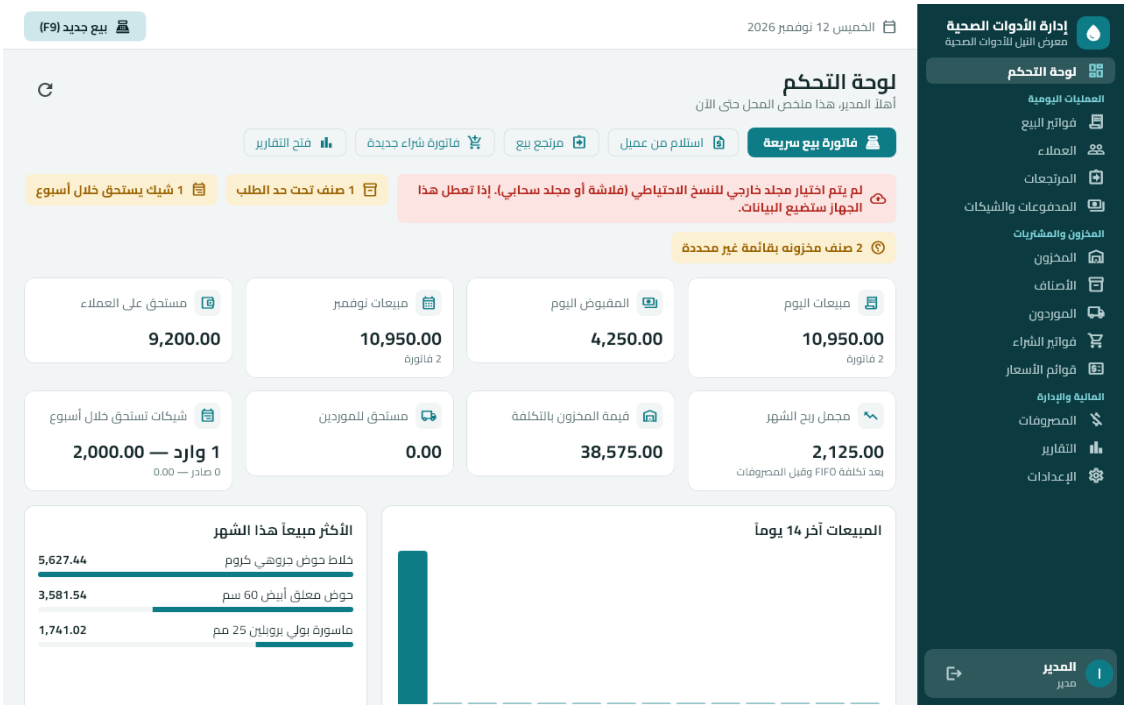
Instalux ERP

Application Guide

Version 1.11.0 | database schema 16 | 2026-11-12



1. What the program is

Instalux ERP is an offline, Arabic, right-to-left Windows program that runs a sanitaryware supply business on one PC: products and price lists, stock in several places with FIFO costing, customers and suppliers, buying, selling, quotations and orders, payments and cheques, returns, expenses, reports and backup. This guide describes version 1.11.0 (database schema 16).

Built for one company, one PC and one currency (Egyptian pounds). There is no internet, server or cloud account. The data lives in one database file on that PC, and backups are copied to a second folder.

Rules that keep the numbers right

Rule	What it means
Money is whole piastres	1 EGP = 100 piastres, stored as integers. No decimals, so no rounding drift.
Quantities are thousandths	1.5 metres is stored as 1500. Pieces must be whole; metres may have decimals.
Percentages are basis points	21% is 2100. A discount takes price x bp / 10000, rounded half up.
Stock is never typed in	Stock is the sum of stock movements, per place. Every document adds movements.
Cost comes from FIFO lots	Each receipt is a lot with its actual cost. Sales use the oldest lots first.
Records are never deleted	A wrong document is cancelled with a reason and stays on record.
One transaction per change	Anything touching money or stock is saved completely or not at all.
Nothing is invented	A missing price, discount, VAT rate or cost stays empty until a person supplies it.

What the program can do

- Products as master data only: fixed Product ID, maker code kept as text, brand, category and subcategory, series, variant, finish, dimensions, unit, barcode, specs, verified VAT basis and rate, and a photo (shrunk to a small JPEG and kept inside the database, so backups carry it; thumbnails in the lists).
- Price lists as dated versions (Price List ID, brand, list type, period, source, VAT status), with full Excel import, single or small-group entry, a review queue, and price history per product. An "Arabic name" column names new products in Arabic (the English kept as description) and fills the Arabic name of products named only in English; an Arabic name already there is never replaced.
- Buying policies (supplier + brand + list) apart from selling discounts by customer category (retail, wholesale, projects, engineers, or any the owner adds), with optional special discounts by brand, category or product. Product before category before all.

- Stock in several places (main warehouse, showroom storage, display) with transfers that keep cost and date, reservations, and display-only places that are never sold from.
- Stock at the old and the new price: every lot remembers the list version it came in under.
- Sales invoices with a review before approval (list price, each discount and source, VAT, FIFO cost, profit, margin, markup), quotations and sales orders with a price lock, delivered in parts.
- Purchase invoices and purchase orders with a cost lock, received in parts into a chosen place.
- Returns both ways, payments and cheques, advances on orders, account statements, expenses.
- Fourteen reports with Excel and PDF output, a dashboard, approvals log, audit trail with before and after values.
- Users with two roles, automatic backups with a second folder, restore, Excel import of products, customers, suppliers.

2. Users, roles and sign-in

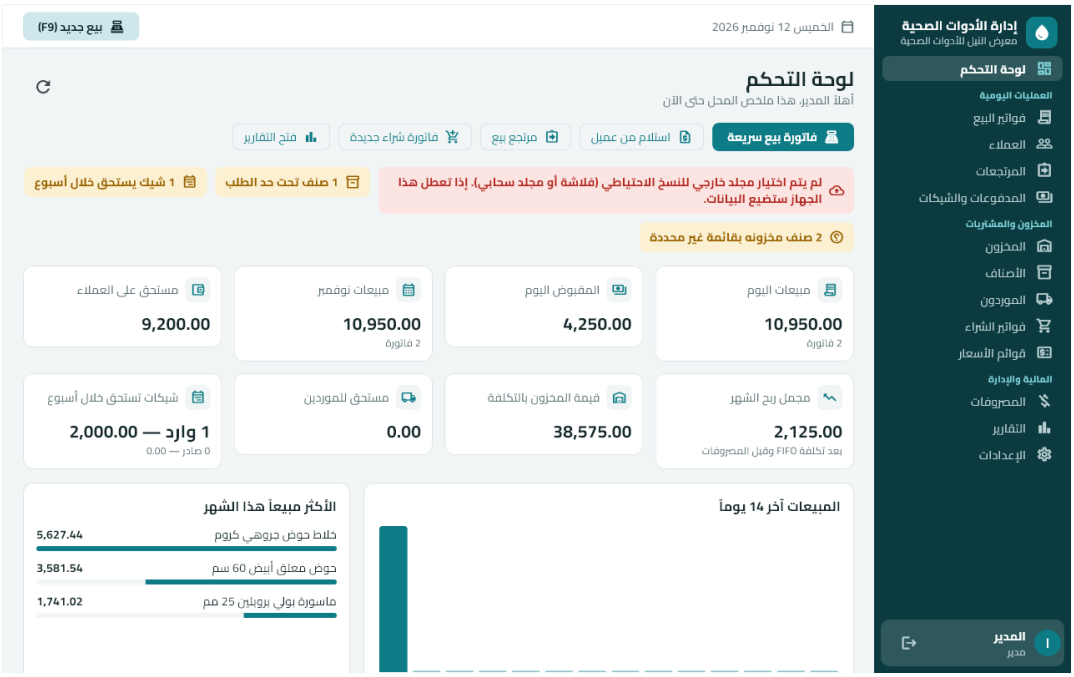
The first start shows a setup screen: company name, address and phone, then the first administrator. After that every start shows the login screen. Passwords are stored only as salted PBKDF2 hashes; every login, failed login and change is written to the audit trail.

Area	Administrator	Cashier
Dashboard, customers, sales, returns, payments	Yes	Yes
Inventory	Adjust stock, reservations, places, value at cost	View and stock card
Products (catalogue)	Create and edit	View
Suppliers, purchases, purchase orders, price lists, policies	Yes	No
Expenses, reports, settings, users, backup, import	Yes	No
Price not from an approved list	Allowed, logged	Refused
Discount past the policy limit, margin under the minimum	Allowed, logged	Refused
Sale below FIFO cost	With a written reason, logged	Refused
Customer over the credit limit	Allowed, logged	Refused
Cost, profit and margin on screen	Shown	Hidden

A cashier who types an admin address is sent back to the dashboard. If nobody can sign in, the developer runs reset_password.bat with the program closed: it backs up the database, sets a new password and re-activates the user.

3. The main window

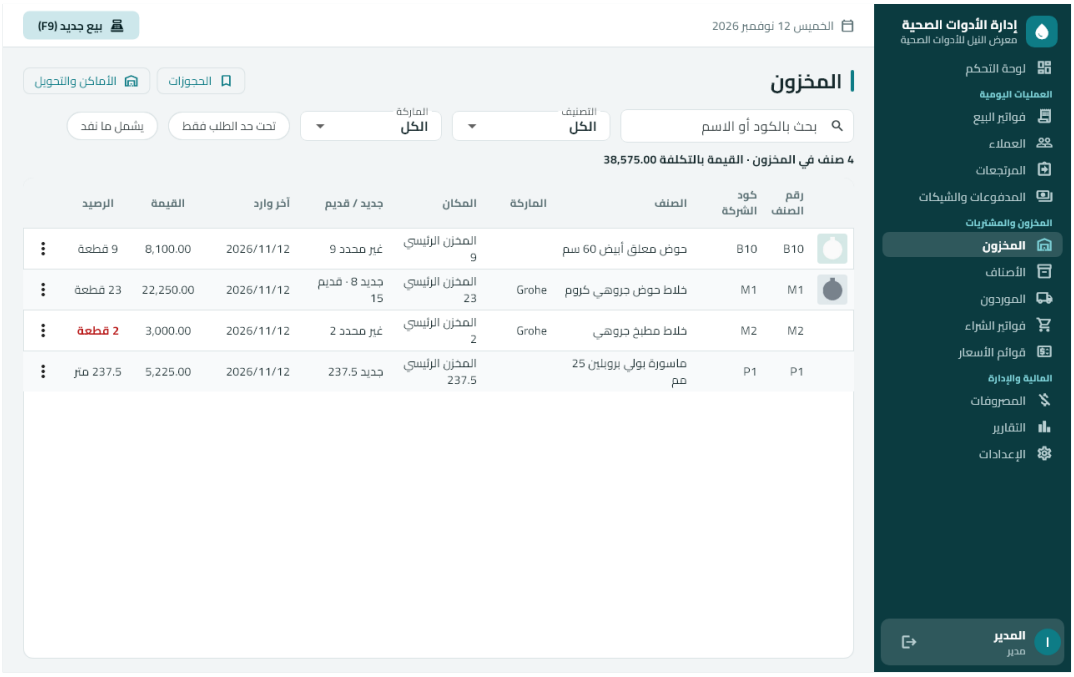
A dark sidebar on the right lists the screens in three groups: daily work (sales, customers, returns, payments), stock and buying (inventory, products, suppliers, purchases, price lists) and finance and admin (expenses, reports, settings). The top bar shows the date and a "new sale" button; F9 opens a new sale from anywhere.



Dashboard

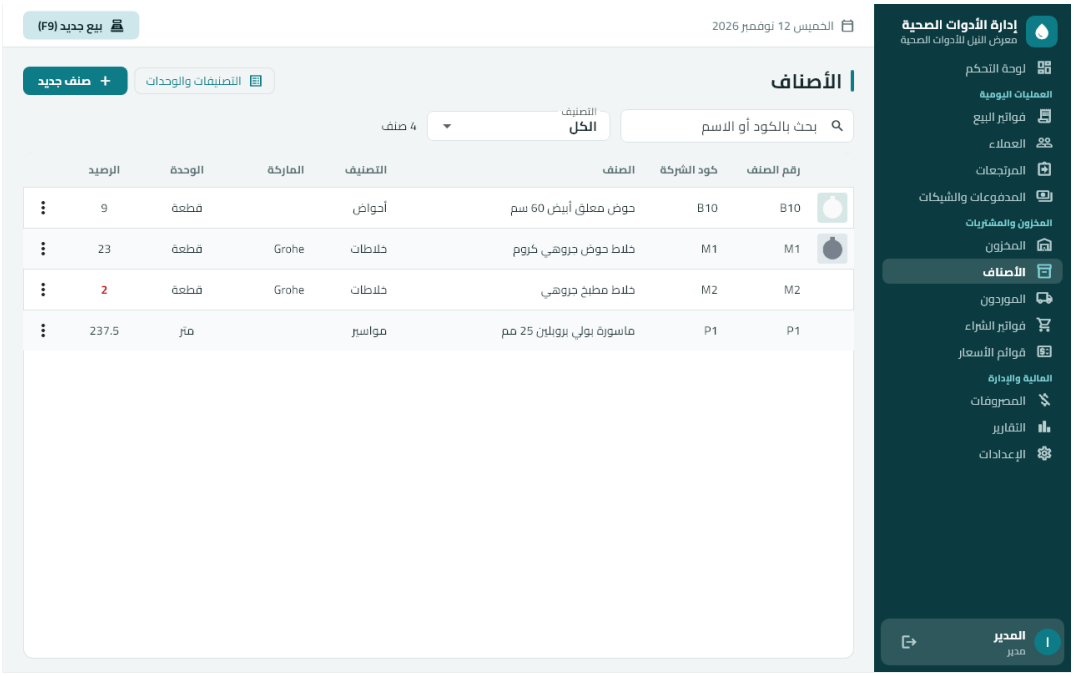
Dashboard part	What it shows
Quick actions	Fast sale, receive from a customer, sale return; for the admin a purchase invoice and reports.
Attention strip	Missing outside backup, items under minimum, cheques due within a week, price lists waiting for review, stopped price rows, stock whose price list is unknown. Each opens its screen.
Figure cards	Today's sales and money received, this month's sales against last month, what customers owe; for the admin also gross profit this month, stock value at FIFO cost and what is owed to suppliers.
Charts and lists	Sales for the last 14 days, best sellers this month, items under minimum, top debtors.

4. Inventory, products and places



Inventory: only the stock the shop holds

The Inventory screen lists only products that came into stock (a purchase, opening stock, an adjustment or a return), never the whole catalogue loaded from price lists. Each row shows the balance, where it is, how much is at the new and the old list price, the last receipt and, for the admin, the value at FIFO cost; the total value is at the top. Filters: search, category, brand, under minimum, and "include sold out". Reservations and places and transfers open from this screen.



Products: the catalogue, including everything loaded from price lists

- Product ID: given automatically when left blank (P-000001 ...) and never changes. The maker's code is kept exactly as text. The same maker's code under the same brand and variant is refused as a duplicate.
- The match key used for search and duplicates ignores spaces, letter case and Arabic digits, but keeps leading zeros and hyphens: "0123" is not "123".
- The product holds no price, cost or balance. Prices live in price list versions, cost in FIFO lots.
- Opening stock needs quantity, actual cost and place, and makes a movement and a FIFO lot.
- Places: the main warehouse (always there) and others with a kind, a sellable flag and an active flag. Display-only places never count as available to sell.
- Transfers move stock with its original cost, received date and price list version.
- Stock adjustment: per place, with a reason, at the last known cost.
- Reservations hold stock for a customer; nobody else can buy it, and the customer's own purchase uses it up. They are not typed by hand: confirming a sales order holds what is in stock; what is short can be put on a draft purchase order to a supplier, tied to that customer order, and is held for the customer when it is received. Delivering a line in full or cancelling the order releases it. The admin can still release one.



A sales order: held in stock, on order from a supplier, short

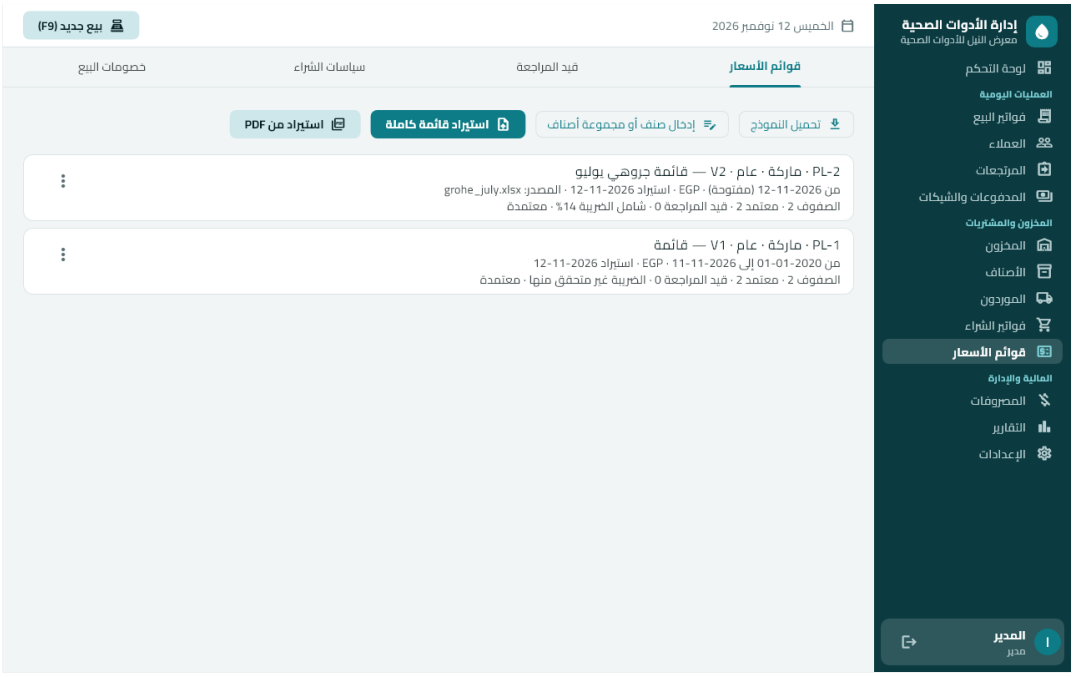
Stock at the old and the new price

Every lot remembers the price list version it came in under and that version's list price. Stock counts as new when its list price equals today's price, old when the price has changed since, and unknown when it came in before any list existed. Unknown is never guessed: an admin assigns it per product, or for a whole list at go-live. Selling always takes the oldest stock first.



Stock by price for one product

5. Price lists and policies

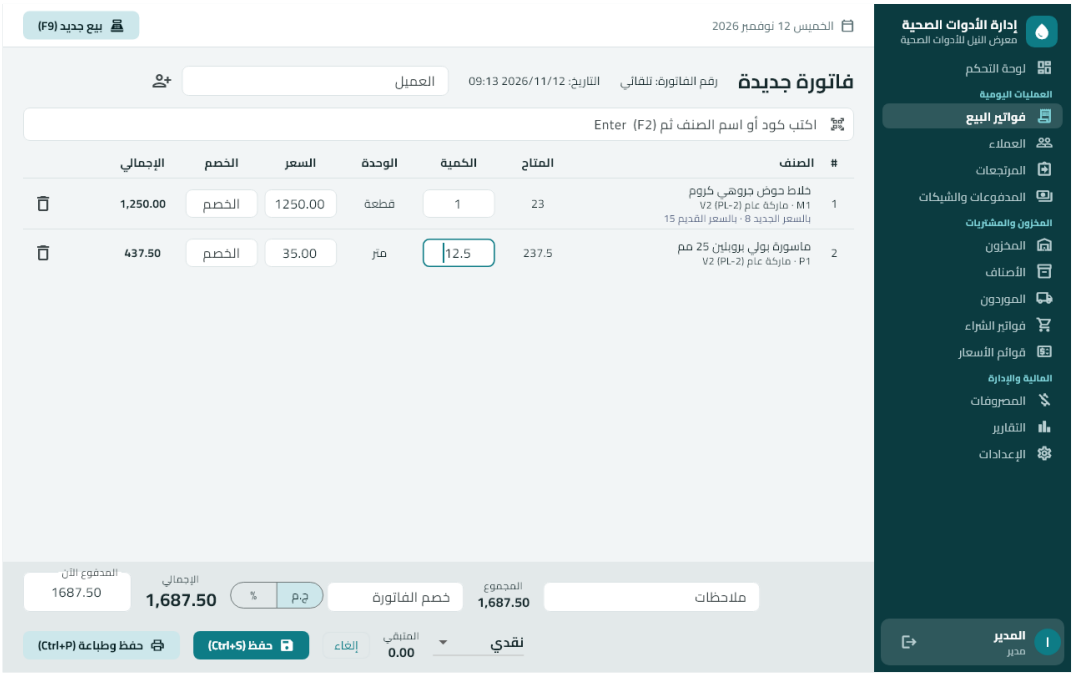


Price list versions

Item	Behaviour
Version	Each list file is a new version with its own ID (PL-n), brand, list type, version number, currency (EGP), effective from and to, import date, source file and pages, row count and VAT status.
Period	A new full version ends the previous one the day before it starts; one sent again on the same day replaces it from that day. A version for a few products leaves the earlier list open for everything else.
Same type, new version	Only the same brand and list type make a new version. The import shows the brand's list types to pick from and warns in red when a new type would make a separate list. A new version takes over the active buying policies of the one it follows, with the same discounts. The format to prepare is in docs/price_list_format_ar.pdf, with two templates in docs/templates.
Never overwritten	A product priced in a version cannot be changed in it. A mistaken version is withdrawn, never deleted, and documents keep using what they used.
Full import	Excel with a name and a price column, and the maker's code when the list has one. Known products get a price, new ones a Product ID, duplicates and ambiguous rows stop in the review queue. A list with no codes matches by exact name within the brand and colour.

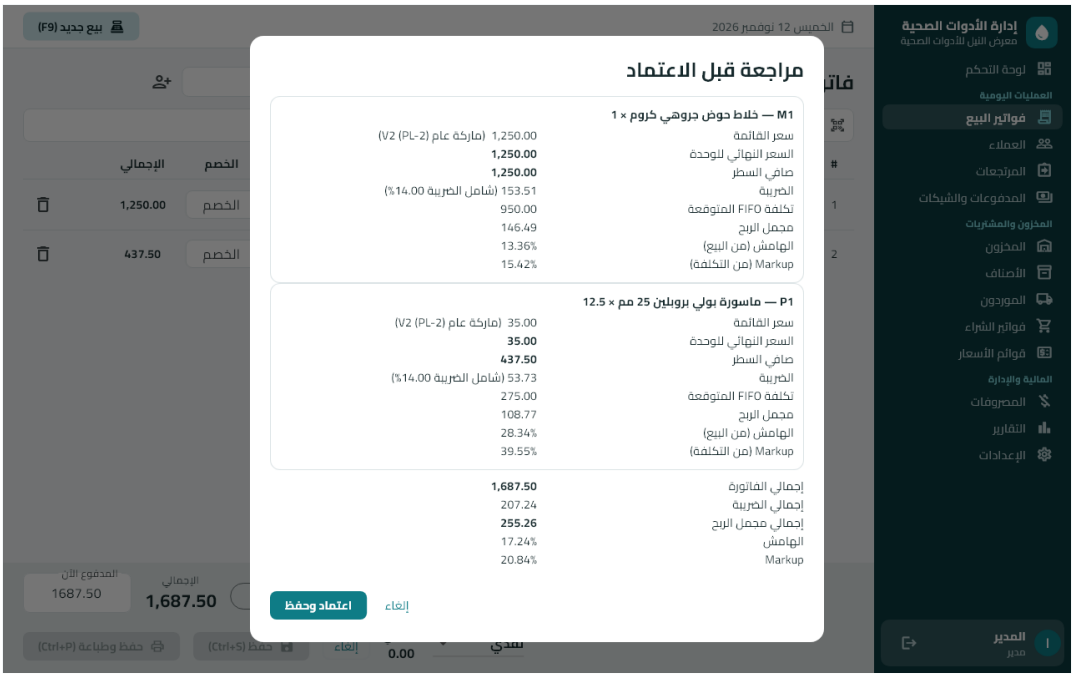
Item	Behaviour
PDF import	A supplier PDF with real text is read page by page: codes, prices and the English description. Price columns come from the headers (White, Pergamon, Colours, Price) or from where prices line up; the person ticks the columns to take and names each one by its colour. A number under no column stops the row for review with its page. Scanned pages (pictures) are refused, never guessed. The PDF is kept with the version.
Small entry	A product or a few, into an existing version or a new dated one, with no file.
Review queue	Link a stopped row to a product, make a new product, or skip it with a reason.
Buying policy	Supplier + brand + list, scope all, category or product, base and extra discount. Without a policy no discount is assumed and the cost is typed.
Customer category	Retail, wholesale, projects and contractors, engineers, and any the owner adds, each with a usual discount. Every customer has one; a walk-in sale is retail. Not tied to a list version.
Special selling discount	Customer category + brand (or every brand), scope all, category or product, a discount that replaces the usual one, optional maximum discount and minimum margin. The brand rule beats the every-brand rule at the same scope. The customer's own discount and the negotiation add on top: 21% + 1 point = 22%.

6. Selling



Fast sale screen

- Type a code, name or barcode and press Enter; the quantity takes focus. Ctrl+S saves, Ctrl+P saves and prints.
- Each line starts at the approved list price with the policy and customer discount filled in. When several lists can price a product the cashier chooses one; with none, only an admin can type a price.
- The invoice discount is typed as an amount or a percent (a switch beside it) and is split over the lines exactly, for profit and tax. Purchases work the same way.
- Stock comes from sellable places, main warehouse first, or from the place chosen on the line.
- Before saving, a review shows everything about the price and, for the admin, cost and profit.

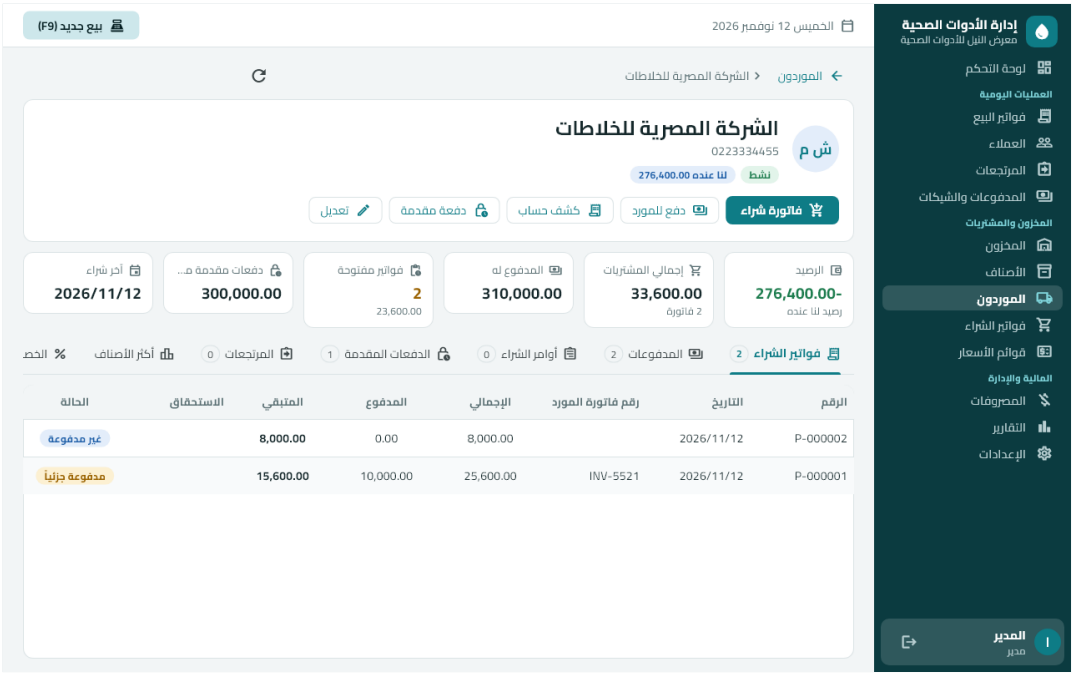


Review before approval

Quotations and sales orders: draft or quotation, confirm (shows the review and locks the version, list price, discounts and VAT status), then deliver in parts; each delivery is an ordinary invoice at the locked price. An order approved on V1 keeps V1 after V2 starts. Advances received on an order move to the invoice. An outside order number can be used once per channel.

Record pages and document views

- Clicking a supplier, customer or product in any list opens its page: a header with its actions (a new invoice for that party, a payment, the statement, editing), key figures, and tabs with every invoice, payment, order, advance, return, reservation, movement and price linked to it.
- Sales, purchases, payments, returns and expenses lists show the period's figures on top and filter chips with how many invoices are paid, partly paid, unpaid, overdue or cancelled.
- A posted invoice opens as a document: a status bar, the party as a link to its page, the lines, the totals and the payments made on it. Sales and purchase orders have screens of their own, with what is still to deliver or receive.

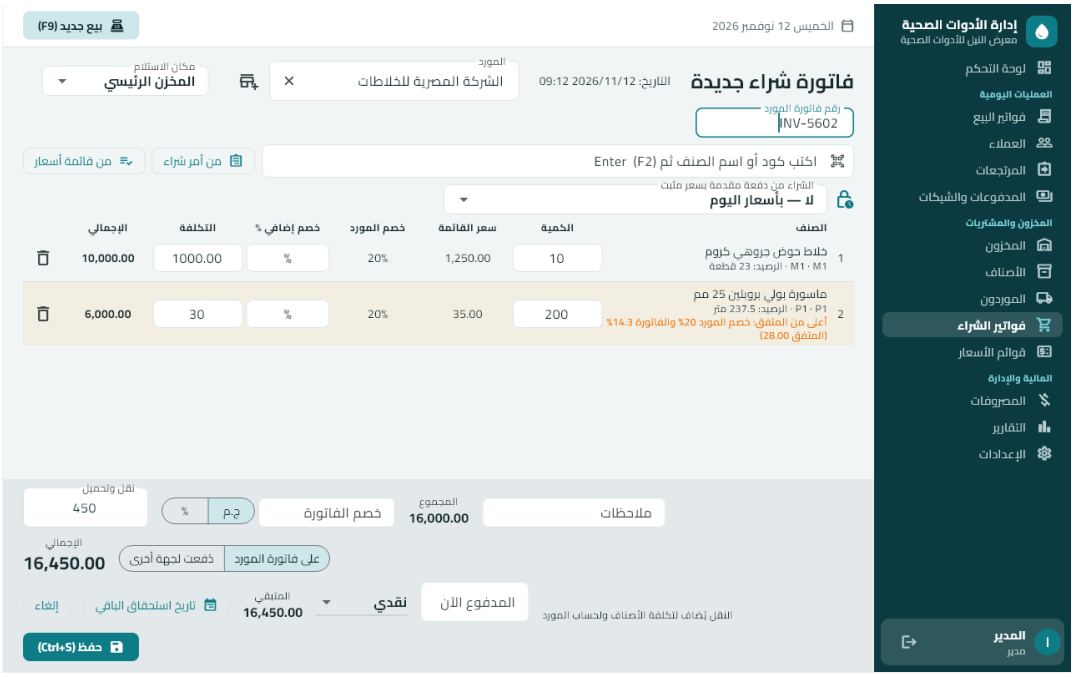


A supplier's page

7. Buying

- Purchase invoice (admin): supplier, their invoice number, receiving place, lines, invoice discount, transport, paid now. Each line shows the list price, the supplier's agreed discount, an extra % for that line, and the final cost. A cost above the agreed one is marked orange and asked about before saving.
- Many lines at once: "from a purchase order" fills the order's remaining lines at their locked cost (other products can be added to the same invoice); "from a price list" ticks products of a list with quantities.
- Each line makes a FIFO lot at its net cost: the invoice discount is split over the lines and lowers the cost; transport and unloading are split by value and raise it. On the supplier's invoice they are owed to the supplier; paid to someone else they become a purchase expense that net profit does not take off twice.

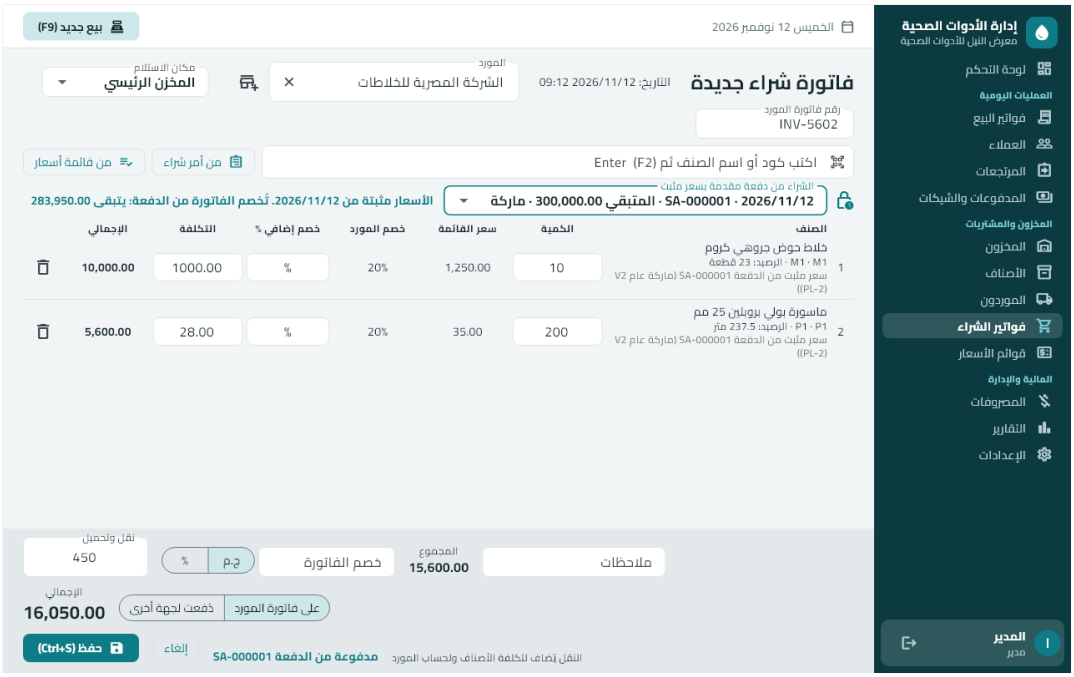
- The supplier's invoice number used twice for the same supplier is asked about. Paid now: cash, bank or a cheque. What is left can have a due date; the dashboard lists supplier invoices due within a week and those overdue.
- Purchase orders: draft, approve (locks list, policy and cost, or takes a typed cost), receive in parts into a chosen place; each receipt is an ordinary purchase invoice. Advances to suppliers move to the invoice.
- Cancel (admin, with a reason): empties the lots it made, place by place; refused if the goods were sold. Its cheque and its transport expense are cancelled with it.



Purchase invoice: list price, agreed discount, extra %, transport

Supplier advance locked to a price list

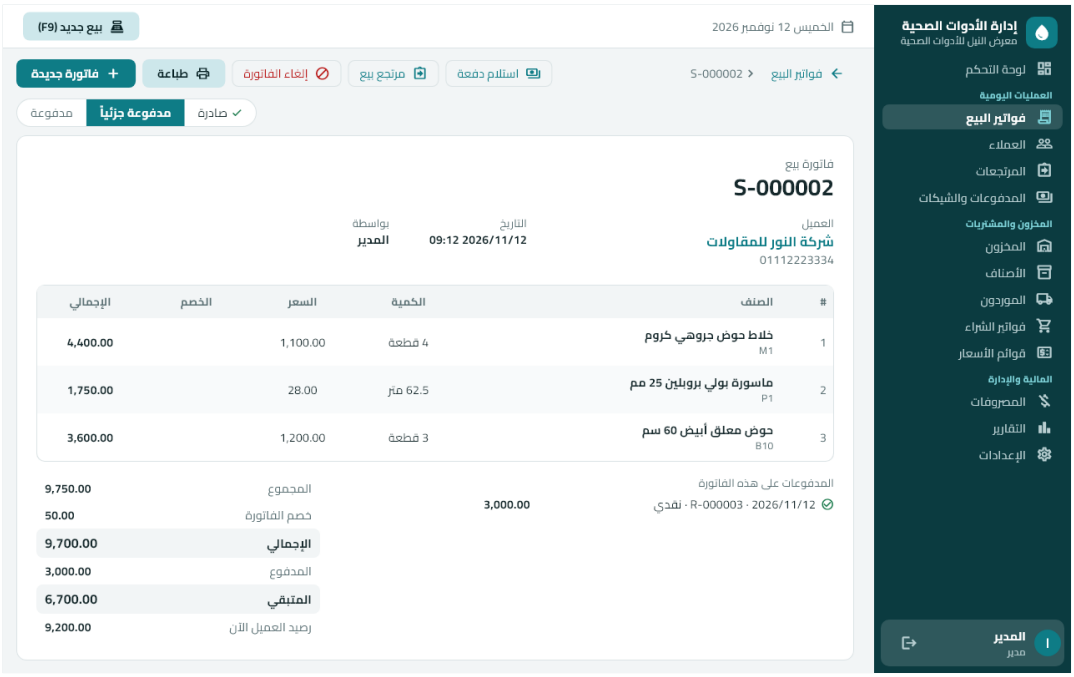
- Money paid to a supplier ahead (e.g. 300,000), from the supplier's menu. It locks the price lists in force that day for the brands chosen, and the supplier's discount on them that day; not particular products.
- It counts in the supplier's balance at once. Each advance shows what has been taken and what is left.
- A purchase invoice for that supplier can be "from the advance": every line takes its locked list price and discount, whatever newer lists or discounts exist, and the invoice is paid from the advance. An invoice bigger than what is left, or a product not on the locked lists, is refused.
- Cancelling such an invoice gives its value back to the advance. An advance can be cancelled only while nothing has been taken; its payment cannot be cancelled from the payments screen.



A purchase taken against an advance at the locked prices

8. Returns, payments, balances and expenses

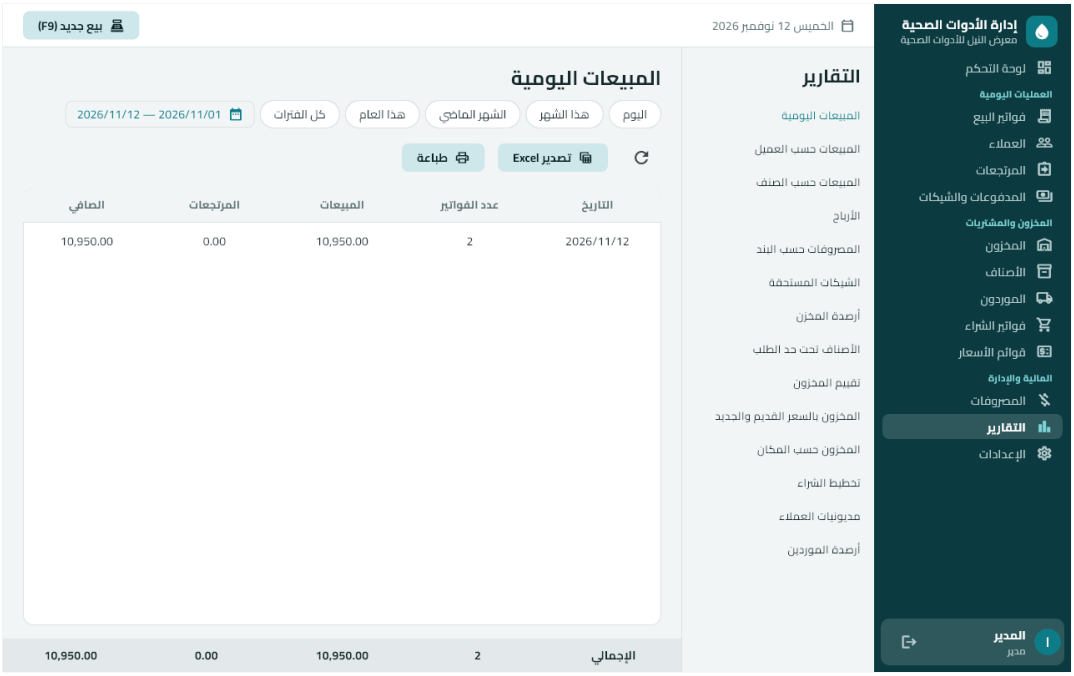
- Sale return (anyone) back into a chosen place, keeping the price list version the goods left under; purchase return (admin) leaving at that purchase's own cost. Refunds follow the net after the invoice discount.
- Payments: received from customers, paid to suppliers (admin); cash, bank or cheque; linked to an invoice or on account. Cheques: pending, cleared, bounced or cancelled.
- Balances are always calculated from documents: opening balance + invoices - returns + refunds - payments.
- Statements for any date range with a running balance; uncleared cheques listed apart.
- Expenses by category with date and note; cancelled with a reason, never deleted.



An invoice: print, take a payment, return, cancel

9. Reports

Report	Answers
Daily sales	What was sold each day
Sales by customer	Who bought how much
Sales by product	Quantity and value per item, after the invoice discount share
Profit	Net sales, exact FIFO cost of goods sold, gross profit, expenses, net profit
Expenses by category	Where the money went
Cheques due	Cheques in and out falling due
Stock levels / Items under minimum	Quantity now; what to reorder
Stock valuation	Value of stock on hand at FIFO cost
Stock at old and new price	Per product: version, list price, current price, quantity, actual cost
Stock by place	Per place, with what is display only, and its value
Purchase plan	Minimum plus reserved less stock and open orders, with the last supplier
Customers owing / Supplier balances	Who owes whom



Reports: on screen, to Excel, to PDF

10. Backup, settings and safety

- Backups: on demand, on closing the program, and before a restore or upgrade. The newest 10 are kept, plus a copy in an optional second folder (USB or synced). The dashboard warns when that copy is missing or old.
- Restore checks the zip, swaps it in at the next start, and backs up the current data first.
- Settings: company details and logo, negative stock, pending cheques, users, Excel import, audit trail, printer test.
- Audit trail: every create, update (with before and after for products, customers and suppliers), cancel, login, backup and restore. Approvals log: price override, extra discount, low margin, below cost, credit limit.
- Data folder: %APPDATA%\SanitarywareERP (database, backups, logs\errors.log, logo). Only one copy runs at a time.

11. What this version does not include

Measured against the Instalux ERP Business Blueprint, these are not built yet and must be agreed in writing at delivery:

- Double-entry accounting: chart of accounts, journal entries, trial balance, balance sheet, cash flow, VAT report.
- VAT added to invoice totals (it is verified per list, shown and stored per line).
- Purchase requisitions, supplier comparison, goods receipts with accepted and rejected quantities.
- Customer and supplier aging, due dates and payment terms.
- Month closing and reopening; roles beyond Admin and Cashier.
- Projects and BOQ; CRM leads; quotation validity and revisions; separate delivery notes and shipping.
- Supplier Direct and Mixed Source sales; In Transit stock; numbered stock counts.
- Shopify connection.

12. Technical reference

Item	Detail
Stack	Flutter for Windows, Dart, drift on SQLite, Riverpod, go_router, pdf, printing, excel, Inno Setup.
Database	Schema 16, upgraded step by step from every older version, each step tested on a copy of the old shape.
Numbers	S- sale, P- purchase, R- payment, T- return, PO- purchase order, SO- sales order, PL- price list.
Tests	Automated tests cover every service and the main screens, including the blueprint's worked examples, a hand-calculated acceptance script and a one-year performance test.
Release	build_release.bat runs every test, builds, and makes dist\SanitarywareERP-Setup-<version>.exe. See docs\RELEASE_CHECKLIST.md.